

Elasticity and Fluid Mechanics, **solved**

Step-by-step solutions to every practice problem in Chapter 10. Try each problem on your own before you read its solution.

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P10.1

A pump at the top of the pipe cannot “pull” water up. It can only remove the air above the water. It is the **atmosphere**, pressing on the open water surface at the bottom, that pushes the water up the pipe. The water stops rising when the weight of the column balances the atmospheric pressure, that is, when the pressure at the top of the column has fallen to zero:

$$h_{\max} = \frac{p_{\text{atm}}}{\rho g} = \frac{1.013 \times 10^5}{1000(9.8)} = 10.3 \text{ m.}$$

In practice the limit is a little lower, because water boils at low pressure (its vapour pressure is about 2.3 kPa at 20 °C). Deep wells therefore use submerged pumps that *push* water up from below.

P10.2

The water level falls.

- **Anchor in the boat.** The floating boat displaces a weight of water equal to its total weight, including the anchor. So the anchor accounts for a displaced volume m/ρ_w .
- **Anchor on the bottom.** The anchor now displaces only its own volume, m/ρ_{iron} .

Since $\rho_{\text{iron}} \approx 7.9\rho_w$, the second volume is about eight times smaller, so less water is displaced and the level drops.

P10.3

The continuity equation applies to the **total** flow, $Q = \sum A_i v_i$. The blood from the single aorta is shared among billions of capillaries in parallel. Each capillary is tiny, but their combined cross-sectional area (about 0.18 m^2) is roughly 600 times that of the aorta (3 cm^2). With 600 times the total area, the average speed is 600 times lower. “Narrower is faster” holds only for a single tube carrying all of the flow.

P10.4

Blood pressure varies with height by ρgh , about 78 mmHg per metre (Example 10.3). The pressure that matters clinically is the pressure the heart produces, so the cuff must be level with the heart. With the cuff 30 cm *below* heart level, the reading is too high by

$$\Delta p = 78 \text{ mmHg/m} \times 0.30 \text{ m} = 23 \text{ mmHg}.$$

A normal 120/80 would read about 143/103, falsely suggesting hypertension.

P10.5

Maximum cable tension, during the upward acceleration:

$$T = m(g + a) = 1000(9.8 + 1.5) = 1.13 \times 10^4 \text{ N}.$$

Allowed stress: $250/5 = 50 \text{ MPa}$. So the cross-sectional area must be at least

$$A = \frac{T}{\sigma_{\text{allow}}} = \frac{1.13 \times 10^4}{5.0 \times 10^7} = 2.26 \times 10^{-4} \text{ m}^2.$$

$$d = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(2.26 \times 10^{-4})}{\pi}} = 0.017 \text{ m} = 1.7 \text{ cm}.$$

P10.6

(a) **Fracture force:**

$$F = \sigma A = (1.7 \times 10^8)(3.0 \times 10^{-4}) = 5.1 \times 10^4 \text{ N}.$$

(b) **Comparison.** In Example 6.6, the stiff-legged landing produced about 6.9×10^4 N in total, so about 3.5×10^4 N per leg. That is roughly 70% of the pure-compression fracture load. In a real landing, the load is not perfectly axial. Bending stresses add to the compression, and bone is weaker in tension and bending, so the actual margin is even smaller. A stiff-legged landing from about a metre is a genuine fracture risk, particularly for older adults with lower bone density. Bending the knees cuts the force about 30-fold.

P10.7

(a) **Force on the small piston:**

$$F_1 = F_2 \frac{A_1}{A_2} = (1500)(9.8) \frac{5.0}{500} = 147 \text{ N.}$$

That is about the weight of a 15 kg mass, enough to lift a car.

(b) **Distance.** The volume of fluid moved is the same at both pistons:

$$d_1 = d_2 \frac{A_2}{A_1} = (0.10 \text{ m})(100) = 10 \text{ m.}$$

This is delivered in many short pump strokes. **Energy check:** $F_1 d_1 = 147(10) = 1470$ J and $F_2 d_2 = 14\,700(0.10) = 1470$ J. ✓ The press multiplies force, not energy.

P10.8

(a) **Absolute pressure at 30 m:**

$$p = p_{\text{atm}} + \rho gh = 1.013 \times 10^5 + 1025(9.8)(30) = 1.013 \times 10^5 + 3.01 \times 10^5 = 4.0 \times 10^5 \text{ Pa} \approx 4.0 \text{ atm}$$

(b) **Lung volume.** With pV constant:

$$V = 6.0 \text{ L} \times \frac{1.013 \times 10^5}{4.03 \times 10^5} = 1.5 \text{ L.}$$

Her chest is squeezed to a quarter of its surface volume.

(c) **The scuba danger.** The scuba diver's lungs are full at 4 atm (6 L of air at 4 atm). If they hold their breath and rise to the surface, that air tries to expand fourfold, to 24 L, far beyond lung capacity. Even a rise of a few metres with a held breath can over-pressurise and tear the alveoli (pulmonary barotrauma). Air can then enter the bloodstream and cause an arterial gas embolism,

which may lodge in the brain. Divers are trained never to hold their breath on ascent, and to breathe out continuously in an emergency ascent.

P10.9

(a) **Minimum height.** The hydrostatic pressure of the column must at least match the venous pressure:

$$p_v = 15 \text{ mmHg} = 15(133.3) = 2.0 \times 10^3 \text{ Pa.}$$

$$h = \frac{p_v}{\rho g} = \frac{2.0 \times 10^3}{1000(9.8)} = 0.20 \text{ m} = 20 \text{ cm.}$$

(b) **Driving pressure at 1.0 m:**

$$\rho g h = 1000(9.8)(1.0) = 9800 \text{ Pa} = 73.5 \text{ mmHg.}$$

Net driving pressure: $73.5 - 15 = 59$ mmHg. Some of this is used up by viscous losses in the tubing and cannula, as Poiseuille's law shows.

P10.10

(a) **Floating depth.** The fraction submerged is $\rho_{\text{wood}}/\rho_w = 0.60$, so the depth submerged is $0.60 \times 0.20 = 0.12$ m.

(b) **Mass to submerge completely.** The buoyant force on the fully submerged cube must equal the weight of the cube plus the added mass m :

$$\rho_w V g = (\rho_{\text{wood}} V + m) g \Rightarrow m = (\rho_w - \rho_{\text{wood}}) V = (1000 - 600)(0.20)^3 = 400(0.008) = 3.2 \text{ kg.}$$

P10.11

Continuity. Halving the diameter quarters the area:

$$v_2 = v_1 \frac{A_1}{A_2} = 2.0 \times 4 = 8.0 \text{ m/s.}$$

Bernoulli (gauge pressures can be used throughout, since p_{atm} appears on both sides):

$$p_2 = p_1 + \frac{1}{2} \rho (v_1^2 - v_2^2) - \rho g (y_2 - y_1).$$

$$p_2 = 2.00 \times 10^5 + \frac{1}{2}(1000)(4.0 - 64) - 1000(9.8)(5.0) = 2.00 \times 10^5 - 3.0 \times 10^4 - 4.9 \times 10^4 = 1.2$$

The gauge pressure in the narrow section is **121 kPa**.

P10.12

The flow resistance is $R \propto r^{-4}$, and the radius becomes **0.90r**.

(a) $R'/R = 1/0.90^4 = 1/0.656 = 1.52$. The resistance increases by 52%.

(b) At fixed Δp , $Q \propto 1/R$, so $Q'/Q = 0.656$. The flow falls by 34%.

(c) To restore the flow, Δp must increase by the same factor as R : 1.52 times, a 52% rise.

Evaluate. This is how the body redistributes blood. During exercise, arterioles in the muscles dilate while those supplying the gut constrict. It is also why chronically constricted arterioles raise blood pressure: the heart must push harder to maintain the flow.

P10.13

(a) **Total force.** Measure y downward from the water surface. Atmospheric pressure acts on both sides of the dam and cancels, so use the gauge pressure ρgy . A horizontal strip of the face of height dy feels a force $dF = \rho gy W dy$:

$$F = \int_0^H \rho g W y dy = \frac{1}{2} \rho g W H^2.$$

(b) **Line of action.** The torque about the base is found by weighting each strip by its height above the base, $H - y$:

$$\tau = \int_0^H (H - y) \rho g W y dy = \rho g W \left(\frac{H^3}{2} - \frac{H^3}{3} \right) = \frac{\rho g W H^3}{6}.$$

$$h_{\text{eff}} = \frac{\tau}{F} = \frac{\rho g W H^3/6}{\rho g W H^2/2} = \frac{H}{3}.$$

(c) **Numbers:**

$$F = \frac{1}{2}(1000)(9.8)(200)(50)^2 = 2.45 \times 10^9 \text{ N}.$$

That is equivalent to the weight of about 250 000 tonnes.

Why thicker at the bottom? The pressure grows linearly with depth, so the lowest part of the face carries the most load. The force also tries to slide the dam along its base and to topple it about its downstream toe. A gravity dam resists this with its own weight, which is why it is much wider at the base than at the top.

P10.14

(a) **Corrected density.** The volume of water displaced is

$$V_{\text{disp}} = \frac{75.0 - 2.5}{1000} = 0.0725 \text{ m}^3.$$

This includes 1.2 L of residual air in the lungs, which adds buoyancy but has negligible mass. The tissue volume is therefore

$$V_{\text{body}} = 0.0725 - 0.0012 = 0.0713 \text{ m}^3, \quad \rho = \frac{75.0}{0.0713} = 1052 \text{ kg/m}^3 = 1.052 \text{ g/cm}^3.$$

(b) **Siri equation:**

$$\% \text{fat} = \frac{495}{1.052} - 450 = 470.6 - 450 = 20.6\%.$$

(c) **Without the correction:**

$$\rho = \frac{75.0}{0.0725} = 1034 \text{ kg/m}^3, \quad \% \text{fat} = \frac{495}{1.034} - 450 = 28.5\%.$$

Error: ignoring just 1.2 L of air overestimates body fat by about **8 percentage points**. The density difference between fat and lean tissue is small, so small volume errors have large effects. Labs therefore measure residual lung volume, often by gas dilution, for every subject.

P10.15

(a) **Rate of fall.** Torricelli's law gives the outflow speed $v = \sqrt{2gh}$. The volume leaving per second, av , equals the rate at which the tank's volume decreases:

$$A \frac{dh}{dt} = -a\sqrt{2gh} \Rightarrow \frac{dh}{dt} = -\frac{a}{A} \sqrt{2gh}.$$

(b) **Draining time.** Separate the variables:

$$\int_{h_0}^0 \frac{dh}{\sqrt{h}} = -\frac{a}{A} \sqrt{2g} \int_0^t dt \Rightarrow 2\sqrt{h_0} = \frac{a}{A} \sqrt{2g} t \Rightarrow t = \frac{A}{a} \sqrt{\frac{2h_0}{g}}.$$

(c) **Numbers.** $a = \pi(0.010)^2 = 3.14 \times 10^{-4} \text{ m}^2$, so $A/a = 3183$:

$$t = 3183 \sqrt{\frac{2(2.0)}{9.8}} = 3183(0.639) = 2.03 \times 10^3 \text{ s} \approx 34 \text{ min.}$$

Why the second half is slower. As the water level drops, the outflow speed $\sqrt{2gh}$ falls, so draining slows down. The time to fall from h_0 to $h_0/2$ is a fraction $1 - 1/\sqrt{2} = 0.29$ of the total. So the first half of the water leaves in 29% of the time, and the second half takes the remaining 71%.

P10.16

(a) **Speed and pressure drop.** Halving the radius quarters the area, so

$$v_2 = 4v_1 = 1.6 \text{ m/s.}$$

$$\Delta p = \frac{1}{2} \rho (v_2^2 - v_1^2) = \frac{1}{2} (1060) (2.56 - 0.16) = 1.27 \times 10^3 \text{ Pa} \approx 9.5 \text{ mmHg.}$$

(b) **Viscous resistance.** $R \propto r^{-4}$, so a narrowed segment of the same length has $2^4 = 16$ times the resistance.

(c) **Reynolds number.** Upstream:

$$Re_1 = \frac{\rho v D}{\eta} = \frac{1060(0.40)(0.0080)}{3.5 \times 10^{-3}} = 970.$$

In the stenosis, v is 4 times larger and D is halved, so Re doubles: $Re_2 \approx 1940$.

Comment. The flow in the narrowing is close to the transition value. More importantly, the fast jet that emerges into the wider vessel downstream is strongly unstable. Jets and disturbed flows become turbulent at much lower Reynolds numbers than smooth pipe flow. So turbulence downstream of the plaque is likely, and with it an audible **bruit**. Doctors listen for bruits over the carotid arteries as a sign of significant narrowing.

P10.17

(a) **Pressure profile.** Substitute $\rho = pM/(RT)$ into $dp/dz = -\rho g$:

$$\frac{dp}{dz} = -\frac{Mg}{RT}p \Rightarrow p = p_0 e^{-z/H}, \quad H = \frac{RT}{Mg}.$$

$$H = \frac{8.314(273)}{0.029(9.8)} = \frac{2270}{0.284} = 7.99 \times 10^3 \text{ m} \approx 8.0 \text{ km}.$$

(b) **Everest:**

$$\frac{p}{p_0} = e^{-8849/7990} = e^{-1.11} = 0.33.$$

This agrees well with the measured summit pressure of about 0.33 atm (253 mmHg).

(c) **Oxygen partial pressure:**

$$p_{\text{O}_2} = 0.21 \times 0.33 \times 760 = 53 \text{ mmHg},$$

compared with 160 mmHg at sea level. Inside the lungs it is lower still, because the air there is saturated with water vapour (47 mmHg) and diluted by carbon dioxide. So the alveolar oxygen pressure is only about 30–35 mmHg. Blood oxygen saturation falls dangerously, causing severe hypoxia, impaired judgement, and a risk of high-altitude pulmonary and cerebral oedema. That is why most summit climbers carry supplementary oxygen. Only a handful of exceptionally adapted climbers have reached the top without it.

P10.18

(a) **Pressures:**

$$\text{small: } \Delta p = \frac{2(0.070)}{5.0 \times 10^{-5}} = 2800 \text{ Pa}, \quad \text{large: } \Delta p = \frac{2(0.070)}{1.0 \times 10^{-4}} = 1400 \text{ Pa}.$$

The small alveolus needs the **higher** pressure. Both are connected to the same airway, so the gas in the small one, at higher pressure, flows into the large one. The small one then shrinks, its required pressure rises even further, and it empties and **collapses** (atelectasis). This is an unstable runaway.

(b) **Surfactant.** Surfactant molecules spread out over the surface of the fluid film. When an alveolus shrinks, the same molecules are crowded into a smaller area, which lowers the surface tension sharply. If γ falls faster than r , the required pressure $2\gamma/r$ actually *decreases* as the

alveolus shrinks. A shrinking alveolus then stops losing air, and the instability is removed. Surfactant also lowers the overall work of breathing.

(c) **Aneurysm.** For a cylindrical vessel at blood pressure p , the wall must carry a tension $T = pr$ per unit length. If a weak spot in the wall bulges, increasing r locally, the tension it must carry *increases* in proportion. Meanwhile the stretched wall becomes thinner, so the *stress* in it (tension divided by thickness) rises even faster. Higher stress stretches the wall further, and so on. This positive feedback is why aneurysms tend to enlarge over time and eventually risk rupture. It is why large aneurysms are repaired surgically, often by inserting a stent-graft that takes over the load.

All worked examples and practice problems in these notes are original. Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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